

The Grad Cafe dataset has several crucial limitations because it consists of anonymously submitted data items, coming from applicants rather than verified data. Voluntary participation creates selection bias, since individuals who submit results may not accurately represent the overall applicant population. It is likely that applicants with stronger academic profiles, who ultimately receive acceptances, are more motivated to post their results, rather than applicants with weaker academic profiles who receive rejections from universities. Along with this, many submissions on the Grad Cafe website do not contain certain metrics such as GPA, scores, nationality, degree type, etc. There is no way to verify the accuracy of individual entries. These factors reduce reliability of many conclusions drawn from the dataset, making it difficult to generalize graduate school applicants based on this alone.

Several results produced from this assignment were surprising compared to more broad admissions analytics. For example, the average GRE score in this analysis was 165.87, which is much higher than the 2023 example provided, being 157. One potential reason for this being the case is self-selection bias. As mentioned in the previous paragraph, applicants with stronger GRE scores may be more likely to post their results online, while those with average or worse scores may not be inclined to post. Similar conclusions could be drawn with the average GPA of 3.78 and acceptance rate of 36.94%, which may not necessarily represent the entire applicant population well. Missing data, such as non-reported GPA or GRE scores, may also affect the accuracy of results. If stronger applicants (academically) are more likely to share these metrics, the calculated averages will likely be higher than true averages of the applicant population. Overall, our analysis provides insight into trends among Grad Cafe users, however it should not act as a representation of the entire population of graduate applicants due to selection bias, lack of independent verification, and missing data.